Contribution Journal - ITAI-1371-ML-LAB7-TEAM8

Project: Machine Learning Lab – Model Evaluation with Logistic Regression on the Titanic Dataset

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Overview

For this lab, our team built and evaluated a logistic regression model on the Titanic dataset. The goal was to practice model evaluation techniques, specifically the confusion matrix, classification metrics (accuracy, precision, recall, F1-score), and cross-validation. Below is a breakdown of what each team member contributed, described in my own words.

Katherine Delores – Contribution Journal

My main responsibility was documenting the team's process and contributions. I kept a running record of what each member worked on, how we divided the tasks, and how the pieces of the lab came together. I made sure the journal accurately reflected the division of labor and captured the reasoning behind our decisions—for example, why we chose logistic regression as our baseline model and why we used k-fold cross-validation instead of a single train/test split. I also reviewed the final deliverables to confirm that the coding portion and the reflective journal were consistent with the work described here. Writing this journal helped me see how important clear documentation is for keeping a team aligned and for making the project reproducible.

Marisabel Morales – Coding

Marisabel was responsible for the coding portion of the lab, and she handled it end to end. She started by loading the Titanic dataset and doing the necessary preprocessing—handling missing values, encoding categorical variables, and selecting the features we would use for the model. She then split the data into training and test sets and built the logistic regression model using scikit-learn. After fitting the model, she generated the confusion matrix and calculated the classification metrics: accuracy, precision, recall, and F1-score. Finally, she implemented cross-validation to get a more reliable estimate of model performance and compared those results against the single-split evaluation. She also added comments throughout the code and produced the visualizations of the confusion matrix, which made our results much easier to interpret. Her careful preprocessing and clean evaluation pipeline were the backbone of the project.

Joshua Balderas – Reflective Journal

Joshua was responsible for the reflective journal, where he documented the team's learning process and takeaways. He wrote about what went well—such as how smoothly the cross-validation implementation went and how the metrics gave us a clearer picture of model performance than accuracy alone—and what was challenging, including deciding how to handle missing data and understanding why precision and recall can tell different stories. He also reflected on how the confusion matrix helped us see the types of errors the model was making, and what we might try next, like tuning hyperparameters or testing other models for comparison. His reflections tied the technical work back to the learning objectives of the lab and gave the project a thoughtful conclusion.

Summary

Our team divided the work clearly: I handled the contribution journal, Marisabel built and evaluated the model, and Joshua wrote the reflective journal. Each person's work fed into the others'—Marisabel's code and results gave Joshua the material to reflect on, and both gave me the information I needed to document everyone's contributions. Working this way kept us accountable and made sure the final deliverables covered both the technical and reflective sides of the lab.

Note: This journal was written by Katherine Delores based on her own observations of the team's work.